

CNG Cylinder Process — Machine Data Integration

How to send live machine data to the JP Minda CNG platform

Prepared by Deepnap Softech · for Jay Fe Cylinders / JP Minda Group · v1.0

1. Overview

Each machine on the CNG line sends its readings to **one secure HTTPS endpoint** as JSON, over a simple REST call (`POST`). That is the entire integration. You do not need to match a fixed schema, share any database credentials, or open inbound access to your network — your machines (or a gateway PC) simply *push* data out to us.

Accept-everything You choose the field names. We accept any shape and store it as-is. New or changed fields never break anything.

2. Endpoint & authentication

Item	Value
URL	<code>POST https://digitisationapi.jpmgroun.co.in/api/v1/cng/ingest</code> (final URL confirmed by Deepnap Softech)
Method	<code>POST</code>
Auth header	<code>x-api-key: <the key we issue you></code>
Content-Type	<code>application/json</code>

3. Request body

Field	Required	Meaning
<code>machineId</code>	required	A unique id for the machine (any naming — e.g. <code>HARDNESS-01</code> or <code>ekc_bottom_milling_01</code>).
<code>timestamp</code>	required	When the reading was taken (ISO-8601 like <code>2026-06-29T10:15:00Z</code> , or an epoch number).
<code>pipeId</code>	optional	The cylinder's Pipe ID / barcode, <i>if the machine knows which cylinder it is measuring</i> (see §6).
<code>data</code>	required	An object of the actual readings. You decide the field names inside here.

Naming is flexible. Both `machineId` / `timestamp` (camelCase) and `machine_id` / `ts` (snake_case) are accepted. Send whatever your PLC already produces.

4. Examples

Two machines, two different field sets — both are accepted without any change on our side:

```
// Hardness tester
{
  "machineId": "HARDNESS-01",
  "timestamp": "2026-06-29T10:15:00Z",
  "pipeId": "PIPE-12345",
  "data": { "hardness": 92, "status": "running" }
}

// Bottom milling machine (snake_case is fine too)
{
  "machine_id": "ekc_bottom_milling_01",
  "ts": "2026-06-29T10:15:01Z",
  "pipe_id": "PIPE-12346",
  "data": { "depth_of_cutting": 5.1, "depth_actual": 5.0, "servo_slow": 14 }
}
```

5. How often to send

One request **per machine**, every **5–15 seconds** (or on each change of reading). Each request is one snapshot of that machine's current values.

6. Linking a reading to a cylinder (Pipe ID)

Telemetry carries a *machine* id but usually not a *cylinder* id. So we need ONE of these — please confirm which your line can do:

- **Option A — include `pipeId` in the payload:** if the machine reads the cylinder's barcode/serial, send it. Simplest and most accurate.
- **Option B — scan-in:** an operator scans the cylinder onto the machine at each stage; our system then attaches that machine's readings to that cylinder automatically. No change to your payload.

7. Responses

Code	Meaning
202 Accepted	Reading received and stored. This is success.
401 Unauthorized	Missing or wrong <code>x-api-key</code> .
400 Bad Request	The body was not valid JSON.

Any valid JSON with a `machineId` and `data` is stored — even unexpected fields. The line never stalls.

8. Security & what we do NOT need

- Keep the `x-api-key` secret; send only over HTTPS. If a key is ever exposed, tell us — we deactivate it and issue a new one.
- We do **not** need: a fixed data schema, database credentials, or any inbound access into your plant network. Your side only makes an outbound HTTPS call.

9. Quick test (one line)

```
curl -X POST https://digitisationapi.jpmpgroup.co.in/api/v1/cng/ingest \
  -H "x-api-key: YOUR_KEY" -H "Content-Type: application/json" \
  -d '{"machineId":"TEST-01","timestamp":"2026-06-29T10:15:00Z","data":{"pressure":301}}'
```

A `202` with `{"ok":true}` confirms your integration works end-to-end.

10. Go-live checklist

- ☐ Deepnap issues you an `x-api-key`.
- ☐ You confirm the Pipe ID method (Option A or B in §6).
- ☐ You send a few test packets; we confirm they arrive (raw + on the dashboard).
- ☐ Point all machines at the endpoint — done.